
Executive Business Report

Sales Forecasting & Demand Analysis Report

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Project: Retail Sales Forecasting using Machine Learning

1. Executive Summary

This report presents an analysis of four years of retail sales data to identify sales trends, forecast future demand, detect unusual sales patterns, and segment products based on demand. Three forecasting techniques were evaluated, and the XGBoost model delivered the best performance with the lowest prediction error. The analysis indicates continued sales growth over the next three months, with strong performance expected in the West region and the Technology category. The findings can support inventory planning, supply chain decisions, and business growth while reducing the risk of stock shortages and excess inventory.

2. Key Findings from Data Analysis & Forecasting

- Sales show an overall upward trend over the four-year period, indicating consistent business growth.
- Seasonal patterns are present, with higher sales during festive and year-end periods.
- Time series decomposition confirmed both long-term growth and recurring seasonal behavior.
- The Augmented Dickey-Fuller (ADF) test indicated that the original sales series was non-stationary. After applying first-order differencing, the series became stationary and suitable for forecasting.
- Three forecasting models were compared:
 - SARIMA
 - Prophet
 - XGBoost

Model Performance

Model	MAE	RMSE	MAPE
SARIMA	18031.40	19009.18	18.97%
Prophet	20250.79	22318.41	21.86%
XGBoost	13915.32	18893.85	13.29%

Based on these results, **XGBoost** is recommended for production because it achieved the lowest forecasting errors.

3. Three-Month Sales Forecast

Using the XGBoost forecasting model, the expected sales for the next three months are:

Month	Forecast Sales
Month 1	51,037.70
Month 2	30,091.78
Month 3	61,376.88

Considering the model's prediction error (MAE $\approx$ **13,915**), the expected sales are likely to fall within approximately **$\pm 14,000$** of these forecast values. This provides a practical confidence range for planning inventory and supply chain activities.

4. Top Three Anomalies Detected

The Isolation Forest algorithm identified **7 unusual sales weeks**. Three important examples include:

Observation	Possible Business Explanation
Very high sales week	Seasonal or festive promotional campaigns increased customer purchases.
Sudden drop in sales	Possible inventory shortages, supply delays, or lower customer demand.
Unexpected sales spike	Large corporate or bulk customer orders increased weekly sales significantly.

The Z-Score method did not detect any anomalies, suggesting that the unusual sales weeks identified by Isolation Forest are likely due to complex patterns rather than simple statistical deviations.

5. Product Demand Segmentation

Products were grouped into four demand segments using K-Means clustering.

Demand Segment	Recommended Stocking Strategy
High Volume, Stable Demand	Maintain higher inventory levels to avoid stock shortages.
Growing Demand	Increase inventory gradually and monitor demand regularly.
High Growth, High Volatility	Maintain safety stock and review inventory frequently.
Low Volume, Stable Demand	Keep lean inventory to reduce storage costs.

The analysis indicates that the **West region** and the **Technology category** are expected to experience the strongest upcoming growth and should receive higher inventory priority.

6. Business Recommendations

Recommendation 1

Increase inventory allocation for the **West region** and **Technology category**, as they show the strongest forecasted growth and are expected to contribute significantly to future sales.

Recommendation 2

Adopt the **XGBoost forecasting model** for monthly demand planning because it achieved the best forecasting accuracy (MAE = **13,915.32**, MAPE = **13.29%**) among the evaluated models.

Recommendation 3

Implement continuous anomaly monitoring to quickly identify unusual sales spikes or declines. Early detection can help improve inventory planning, promotional decisions, and supply chain responsiveness.

7. Risk / Limitation

The forecasting model is based only on historical sales data. Unexpected events such as economic changes, new competitors, policy changes, supply chain disruptions, or major marketing campaigns may affect future sales and reduce forecasting accuracy. Regular model retraining with updated data is recommended to maintain reliable predictions.

Conclusion

This project demonstrates how machine learning and data analytics can support business decision-making by forecasting sales, identifying unusual patterns, and improving inventory planning. The XGBoost model provides the most reliable forecasts, while demand segmentation helps optimize stocking strategies across different product groups. Together, these insights can help improve operational efficiency, reduce inventory costs, and support sustainable business growth.